

Quiz 2

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (8 points) Given the the following matrices:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} -3 & 2 \\ 0 & 1 \end{bmatrix}$$

(a) Compute $\mathbf{AB}$.

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

(b) Compute $\mathbf{CA}$.

$$\mathbf{CA} = \begin{bmatrix} -3 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -7 & -6 & -5 \\ -2 & 0 & 2 \end{bmatrix}$$

(c) Is $\mathbf{AC}$ possible and why?

Not possible since the number of columns of A (3) doesn't equal to the number of rows of C (2)

Problem 2. (6 points)

(a) Find the Cartesian coordinates of the points whose polar coordinates are $(\sqrt{3}, \pi/3)$.

$$\begin{aligned} (r, \theta) &= (\sqrt{3}, \pi/3) \\ \Rightarrow x &= r \cos \theta = \sqrt{3} \cos(\pi/3) = \frac{\sqrt{3}}{2} \\ y &= r \sin \theta = \sqrt{3} \sin(\pi/3) = \sqrt{3} \cdot \frac{\sqrt{3}}{2} = \frac{3}{2} \quad \rightarrow (x, y) = \left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right) \end{aligned}$$

(b) Find the cylindrical coordinates of the points whose Cartesian coordinates are $(-2\sqrt{3}, -2, 2)$.

$$\begin{aligned} (x, y, z) &= (-2\sqrt{3}, -2, 2) \\ \Rightarrow r &= \sqrt{x^2 + y^2} = \sqrt{(-2\sqrt{3})^2 + (-2)^2} = \sqrt{16} = 4 \\ \tan \theta &= \frac{y}{x} = \frac{-2}{(-2\sqrt{3})} = \frac{1}{\sqrt{3}} \quad \left. \begin{array}{l} (-2\sqrt{3}, -2) \text{ is in the third quadrant} \\ z = 2 \end{array} \right\} \Rightarrow \theta = \frac{7}{6}\pi \rightarrow (r, \theta, z) \\ &= (4, \frac{7\pi}{6}, 2) \end{aligned}$$

(c) Find the Cartesian coordinates of the points whose spherical coordinates are $(3, \pi/3, \pi/2)$.

$$\begin{aligned} (\rho, \phi, \theta) &= (3, \pi/3, \pi/2) \\ \Rightarrow x &= \rho \sin \phi \cos \theta = 3 \sin(\frac{\pi}{3}) \cos(\frac{\pi}{2}) = 3 \cdot \frac{\sqrt{3}}{2} \cdot 0 = 0 \\ y &= \rho \sin \phi \sin \theta = 3 \sin(\frac{\pi}{3}) \sin(\frac{\pi}{2}) = 3 \cdot \frac{\sqrt{3}}{2} \cdot 1 = \frac{3\sqrt{3}}{2} \rightarrow (x, y, z) \\ z &= \rho \cos \phi = 3 \cos(\frac{\pi}{3}) = 3 \cdot \frac{1}{2} = \frac{3}{2} \rightarrow (0, \frac{3\sqrt{3}}{2}, \frac{3}{2}) \end{aligned}$$

Problem 3. (6 points) Let

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

Determine whether the function f is continuous at the origin $(0, 0)$ and explain why.

Solution: Along the line $y = mx$ ($x \neq 0$), we have.

$$f(x, y) = f(x, mx) = \frac{2x \cdot mx}{x^2 + (mx)^2} = \frac{2mx^2}{(1+m^2)x^2} = \frac{2m}{1+m^2}$$

$$\begin{aligned} \text{Thus} \\ \lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } y = x}} f(x, y) &= \frac{2 \cdot 1}{1+1^2} = 1 \quad \& \quad \lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } y = -x}} f(x, y) = \frac{2 \cdot (-1)}{1+(-1)^2} = -1 \end{aligned}$$

$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist

$\Rightarrow f(x, y)$ is not continuous at $(0, 0)$.